Marwan Mahmoud Abdelmoniem Zaky

Mechanical Design Engineer

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Summary

Dynamic and detail-oriented Mechanical Design Engineer with extensive academic and hands-on experience in designing and analyzing mechanical systems, automotive components, and aerospace structures. Proficient in advanced CAD tools (SolidWorks, Inventor), Finite Element Analysis (ANSYS, SolidWorks Simulation), and mathematical modeling (MATLAB, Simulink). Successfully contributed to award-winning engineering projects, demonstrating expertise in vehicle dynamics, UAV design, and stress analysis. Adept at developing innovative solutions that enhance performance, reduce costs, and ensure structural integrity. Skilled in collaborating with multidisciplinary teams and delivering projects within stringent deadlines and quality standards.

Core Competencies

- Simulink, MATLAB
- Reverse Engineering(Geomagic Design X)
- Bill of Materials (BOM)
- Model-Based Design (MBD)
- 3D Scanning & 3D Modeling
- Geometrical Dimensioning and Tolerance(GD&T)
- Fluid-Structure Interaction
- Sheet Metal Design
- Aerospace Structures
- Vehicle Dynamics
- Prototype Development
- Microsoft Office
- Finite Element Analysis (FEA)
- Design Optimization
- CAD Software: (SolidWorks, Inventor)
- FEATools:ANSYS(Explicit dynamics, structural optimization, material modeling, Static, Modal Harmonic, Transient, CFD)
- SolidWorks Simulation
- SolidWorks Motion Analysis

Professional Experience

Vehicle Dynamics Member

EVER V (Helwan Racing Team) | Dec 2023 to Aug 2024

- Develop a 7-DOF ride car model using MATLAB Simulink to optimize ride rate and roll rate.
- Conduct dynamic performance simulations to enhance vehicle handling and overall stability.
- Secure first place in all competition categories, including design, cost, endurance, and autocross races.
- Collaborate with a multidisciplinary team to integrate design improvements across the vehicle.
- Present innovative cost vs. performance analysis, earning top recognition for cost-effectiveness.

Aerostructure Team Leader

AIAA (Aero ASU) | Aug 2023 to Apr 2024

- Led the design and analysis of UAV structures adhering to AIAA rules for loads, pressure, and flight dynamics.
- Conducted Fluid-Structure Interaction (FSI) analysis using ANSYS Structure and Fluent for optimized design.
- Developed weight distribution strategies to enhance aerodynamic performance and stability.
- Collaborated with cross-functional teams to meet project deadlines and quality standards.
- Presented final design and analysis results, earning recognition for technical excellence.

Vehicle Dynamics Member

Shell Eco Marathon (Helwan Racing Team) | Mar 2023 to Dec 2023

- Designed and optimized a central pivot steering mechanism with Ackermann kinematics for precision control.
- Conducted stress analysis on steering components using SolidWorks Simulation and ANSYS.
- Produced detailed manufacturing drawings for all steering parts, ensuring seamless production.
- Contributed to achieving first place in Egypt for overall design presentation through innovative solutions.
- Improved vehicle performance by integrating lightweight and cost-effective design elements.

Vehicle Dynamics Member

GEVIC (Helwan Racing Team) | May 2022 to Feb 2023

- Contributed to manufacturing a high-performance formula car for GEVIC Global competition in Dubai.
- Designed and analyzed components to optimize vehicle dynamics and reduce weight.
- Secured first place in design presentation and third place overall through technical innovations.
- Assisted in integrating subsystems to meet strict performance and safety standards.
- Documented design processes and findings to support future team projects.

Vehicle Dynamics Member
EVER IV (ASU Racing Team) | Oct 2021 to Apr 2022

- Designed and analyzed a suspension system tailored to the car’s wheelbase and track width.
- Created a 3D model of the suspension system using Inventor, ensuring precision and reliability.
- Conducted simulations to evaluate stress points and optimize system performance under dynamic loads.
- Collaborated with team members to assemble and test the suspension system on a prototype.
- Enhanced vehicle handling and stability through iterative design improvements.

Maintenance and Operation Engineer Intern
Arab Organization for Industrialization | Aug 2021 to Sep 2021

Maintenance Engineer Intern
Iveco | Jul 2022 to Sep 2022

Projects

Metal Die Forming | Metal Forming Technologies (Jun 2023)
- Designed and optimized a progressive die layout using SolidWorks to enhance production efficiency.
- Conducted stress analysis to ensure the die's durability and operational reliability.

Pitch Mechanism for Wind Turbine | Machine Elements Design Course (Jun 2022)
- Designed and manufactured a power screw-driven pitch mechanism using Inventor to control blade angles.
- Developed and implemented Arduino code to automate pitch adjustments via a potentiometer.

Air Boat Design (Nov 2021)
- Designed mechanical components for an airboat, ensuring stability and performance.
- Reviewed mechanical calculations and conducted system-level testing for operational efficiency.

Fluid Coupler for Torque Transition (Nov 2021)
- Designed a fluid coupler system for torque transmission in mechanical applications.
- Reviewed and validated critical calculations and governing equations for operational accuracy.

Courses & Certifications

- Mold Design | Net3lm.com
- Surface Modeling | Net3lm.com
- Advanced Part Modeling | Net3lm.com
- Sheet Metal Weldment | Net3lm.com
- Sheet Metal Die Design | Net3lm.com
- Logo Press Die Design | Net3lm.com
- Plastic Injection Mold Design | Net3lm.com
- Powermill 3-axis | Net3lm.com
- Powermill 4-axis | Net3lm.com
- CAMWORKS Turning (2-axis) | Net3lm.com
- Geomagic Design X Basics | Net3lm.com
- ICDL Certification | ICDL Arabia (2017)
- SolidWorks Certifications:
- CSWP (Certified SolidWorks Professional) | Dassault Systèmes (2023)
- CSWA (Certified SolidWorks Associate) | Dassault Systèmes (2023)
- SolidWorks Simulation | Net3lm.com
- ANSYS Specializations | Udemy:
- Static Analysis
- Dynamic Analysis
- Material Modeling
- Mesh Optimization (CFD)
- ANSYS Fluent (CFD)
- MATLAB Onramp (MathWorks)
- Simulink Onramp (MathWorks)

Education

Ain Shams University | Expected Jan 2025
Bachelor of Mechanical Design and Production Engineering

Graduation Project: Vibrio-Polishing Machine (Grade: A)
- Designed and manufactured a machine to improve surface quality and material appearance.
- Utilized MATLAB and Simulink for Model-Based Design (MBD) to optimize system performance.
- Conducted detailed Finite Element Analysis (FEA) to ensure structural integrity and reliability.
- Developed a bill of materials (BOM) and manufacturing plans for production scalability.
- Presented project outcomes to faculty and industry professionals, earning distinction.

Language

- **Arabic :** Native
- **English** : Proficient (B2)